WEEK 13 OF 15 · STAGE 3: ADVANCED METHODS

Uncertainty and scenario discovery

One run is one guess; many runs show what matters

New R skill Batch runs; scenario discovery with a classification tree (rpart)

Modelling Climate and Energy Futures · beginner-friendly lecture and lab slides

COURSE MAP

Where we are in the course

Stage 1 **Foundations**	Week 1 Why model the future?	Week 2 Climate primer	Week 3 Energy primer	Week 4 Economics primer	Week 5 Cost-benefit IAMs: DICE	Week 6 The scenario framework
Stage 2 **GCAM in depth**	Week 7 Inside GCAM	Week 8 Energy in GCAM	Week 9 Land, water and bioenergy	Week 10 Emissions and policy		
Stage 3 **Advanced methods**	Week 11 Optimisation IAMs	Week 12 Why models disagree	**Week 13 Uncertainty**	Week 14 Frontiers and limits	Week 15 Projects and review	

TODAY'S GOALS

By the end of today you can

1. Name three kinds of uncertainty and how each is explored
2. Count the runs in an ensemble and explain why they grow so fast
3. Use a classification tree to find which inputs lead to high emissions
4. Explain robust decisions and emulators in plain words

? **Big question** We do not know future solar costs, population or policy. How can a model help us decide anything when its inputs are so uncertain?

Write the scenario after the runs

Usually you invent a scenario and then run the model. Today we run the model dozens of times first, and let the computer tell us which combinations of assumptions lead to trouble.



What would you want to know before trusting a single model run?

SIDE BY SIDE

Three kinds of uncertainty, three ways to explore them

Kind	Example	How to explore it
Inputs	Future solar cost	Sensitivity runs, ensembles (today)
Model structure	Models built differently	Compare models (week 12)
Future society	Population, growth, policy	Scenarios such as the SSPs (week 6)



Your project's sensitivity test usually covers inputs; say what it leaves out

AN EVERYDAY PICTURE

Weather forecasters never trust one run



A weather service runs its forecast model many times with slightly different starting conditions. If 45 of 50 runs bring rain, you take an umbrella. If they split, you know the forecast is shaky. One run alone would hide that.

What it means for models

1. Many runs show how stable a result is
2. An ensemble varies several inputs together
3. But our inputs are chosen, not random draws

Worked example: counting ensemble runs

Problem The lab ensemble varies solar cost (2 values), CO2 storage (2), SSP (3) and carbon price (3). How many runs?

1. Every combination: 2 × 2 × 3 × 3
2. 2 × 2 = 4, then 4 × 3 = 12, then 12 × 3 = 36 runs
3. Add a fifth input with 3 values: 36 × 3 = 108 runs
4. At 2 hours a run, 4 runs at once: 108 × 2 ÷ 4 = 54 hours

Answer

36 runs in the lab; 108 runs and about 54 hours with one more input

HOW IT WORKS

Scenario discovery in four steps

1

Run many cases

Inputs differ: price, solar cost, CO2 storage, SSP

2

Label each run

High emissions in 2050: yes or no?

3

Fit a tree

It splits runs by input values

4

Read the leaves

Which settings lead to trouble?

Bryant and Lempert introduced the idea; Lamontagne et al. (2018) applied it to a large GCAM ensemble

Worked example: growing a tree by hand

Problem Eight runs; three are labelled HIGH. Which input settings explain them?

1. Split on carbon price. Low price: 4 runs, 3 HIGH. High price: 4 runs, 0 HIGH
2. That is a clean split, so carbon price comes first
3. Split the low-price group on solar cost. Solar high: 2 runs, both HIGH
4. Solar low: 2 runs, 1 HIGH
5. Rule: low carbon price AND high solar cost → HIGH
6. The rule is right on 7 of 8 runs = 87.5%

Answer **Low carbon price with high solar cost leads to high emissions (right on 7 of 8)**

Quick check

In the lab, 9 of 36 runs are labelled high. What does that tell you?

A There is a 25% chance of high emissions

B Under the values we chose, a quarter of runs end high

C The other 27 runs are wrong

D High emissions are unlikely

Quick check: answer

In the lab, 9 of 36 runs are labelled high. What does that tell you?

A. There is a 25% chance of high emissions

B. Under the values we chose, a quarter of runs end high ✓

C. The other 27 runs are wrong

D. High emissions are unlikely

SIDE BY SIDE

Robust choice: keep the worst regret small

Strategy	Future 1	Future 2	Future 3	Worst regret
A: rely on one technology	100	110	200	200 − 125 = 75
B: spread across several	115	118	125	115 − 100 = 15
Best in each future	100	110	125	

Cost to meet a target, index. Regret = your cost − the best cost in that future. Illustrative.

KEY IDEA

Two more tools for a very uncertain future

1

Emulators

A fast stand-in for a slow model. Week 2's rule, 0.45 °C per 1,000 GtCO2, is an emulator of full climate models.

2

Robust decision making

Examine a very wide range of futures and look for near-term choices that hold up in nearly all of them.

3

Why they matter

Thousands of runs need cheap runs; good decisions need to survive the futures we did not expect.

WATCH OUT

Common mix-ups

Not quite	Better
9 of 36 runs are high, so there is a 25% chance	The share depends on the values you chose to try. It is not a probability.
The tree shows what causes high emissions in the real world	It describes how this model behaves across these runs, nothing more.
A tree that fits every run perfectly is the best tree	It may just memorise the runs. Keep it small and check its accuracy.

NEW WORDS THIS WEEK

Words to add to the class glossary

Sensitivity analysis Changing one input at a time, holding the rest fixed, to see how far the result moves. You did this with DICE in week 5.

Ensemble A set of model runs that changes several inputs together, running every combination or a sample of them.

Scenario discovery Running many cases first, labelling the outcomes you care about, then letting a computer find which input settings lead to them.

Emulator A fast, simple formula fitted to a slow model's runs, so thousands of cases can be tried in seconds.

Robust decision A choice that works acceptably across many possible futures, instead of being best in one.

Lab 13: an ensemble and scenario discovery

See how a 36-run GCAM ensemble was set up, label the high-emission runs and let a classification tree find which inputs lead to them.

Script: labs/lab13.R

4 steps

Stuck? Read the error aloud, check the slide, then ask

Read the batch file (no code)

1

Open gcam-files/ensemble_batch.xml. Each ComponentSet is one input; GCAM runs every combination: 2 × 2 × 3 × 3 = 36.

Load the results

One row per run. rpart wants text columns as factors, so convert them.

```
library(tidyverse)
library(rpart)
ens <- read_csv("data/ensemble_results.csv") |>
  mutate(across(where(is.character), factor))
glimpse(ens)
count(ens, solar_cost, ccs_storage,
      ssp, carbon_price)   # 36 rows, n = 1
```

Type it rather than paste it, run one line at a time (Ctrl+Enter), and check the output before the next line.

Label the high-emission runs

A run is high if its 2050 CO2 is above a cut-off you choose. Fill GAP 1: 0.75 marks the top quarter.

```
cut_off <- quantile(ens$co2_2050, ___)  # GAP 1
ens <- ens |>
  mutate(high = factor(if_else(
    co2_2050 > cut_off, "high", "other")))
count(ens, high)
```



Fill each ___ gap before running that line. Stuck? The solution file has the complete version.

Fit the tree and read it

The tree splits runs on one input at a time. Fill GAP 2 with "class" so predict returns labels.

```
tree <- rpart(high ~ solar_cost + ccs_storage +
                ssp + carbon_price,
              data = ens, method = "class",
              minsplit = 6)
plot(tree, margin = 0.1)
text(tree, use.n = TRUE, pretty = 0)
ens <- ens |>
  mutate(pred = predict(tree, type = ___))  # GAP 2
mean(ens$pred == ens$high)
```

Did it work?

You should see

- count(ens, high) shows about 9 high and 27 other
- The tree's first split is on an input you can name
- The share of runs the tree gets right is printed

Worksheet questions

1. Read one path from the top of your tree to a high leaf as a sentence.
2. Which input does the tree use first? Is that what you expected, and why?
3. Change the cut-off to 0.5 and fit the tree again. Does the first split stay the same? What does that say about the label?

RECAP

Take-home messages

1. Uncertainty comes from inputs, model structure and the future of society
2. Ensembles vary several inputs together; every new input multiplies the number of runs
3. Scenario discovery labels runs and fits a tree to find the settings that lead to trouble
4. A share of runs is not a probability, and a tree describes the model, not the world
5. Robust choices keep the worst regret small; emulators make many runs affordable

Next: week 14

Week 14 looks at where IAMs are heading and at the main criticisms of them, and turns your draft results into clear figures.

Practise

Interactive course, week 13: try the ensemble explorer and the 5-question quiz.

Readings: Lamontagne et al. (2018), Large ensemble analytic framework for consequence-driven discovery of climate change scenarios, Earth's Future; rpart package documentation; GCAM v8.2 documentation: How to Get Started (batch mode), Shared Socioeconomic Pathways